

WHIRLPOOL CHALLENGE

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Problem introduction

- **Objective:** Identify the optimal Segment average price and Sell Out.
- **Goal:** Predict weekly sales and estimate the most profitable price point.
- **Context:** Based on Whirlpool's internal sales data from 2023–2025 across six product segments.



WHIRLPOOL CHALLENGE

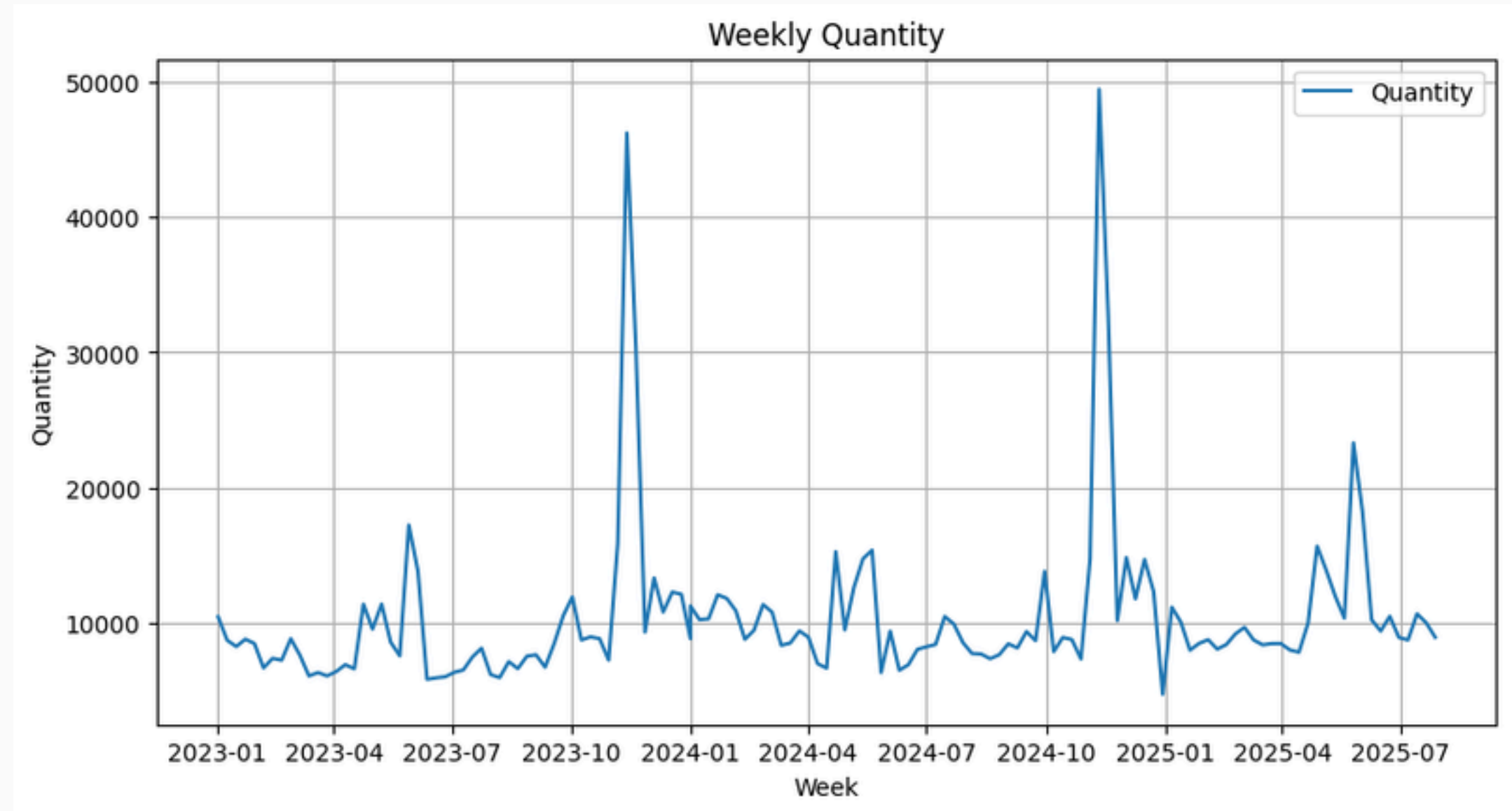
Descriptive Statistics



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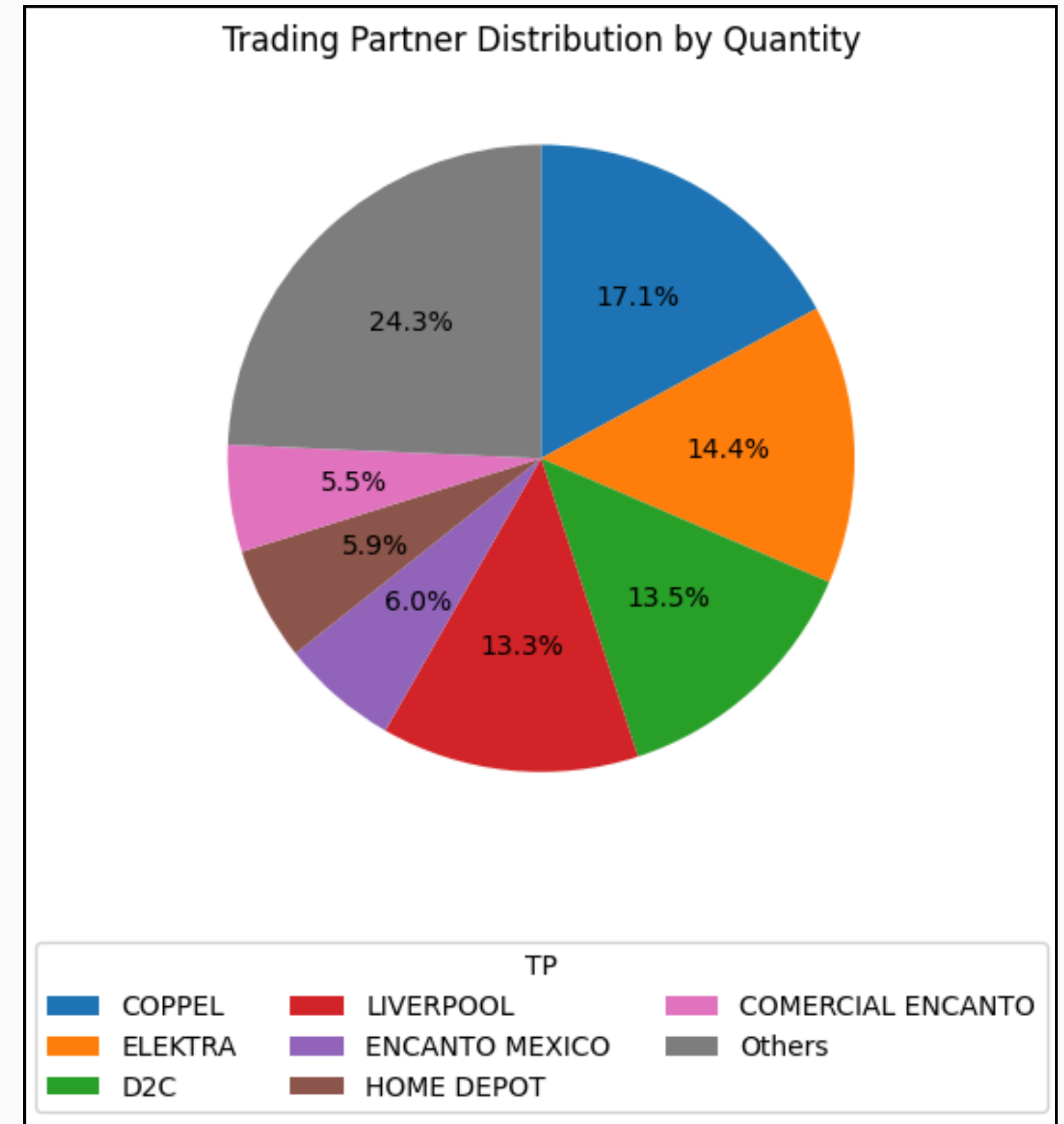
Data description

Each row represents weekly aggregated data (2023–2025) across trade partners and segments; after cleaning NaNs and index columns, the dataset includes 135 aggregated and 30,846 detailed entries with time, pricing, cost, and lagged features.

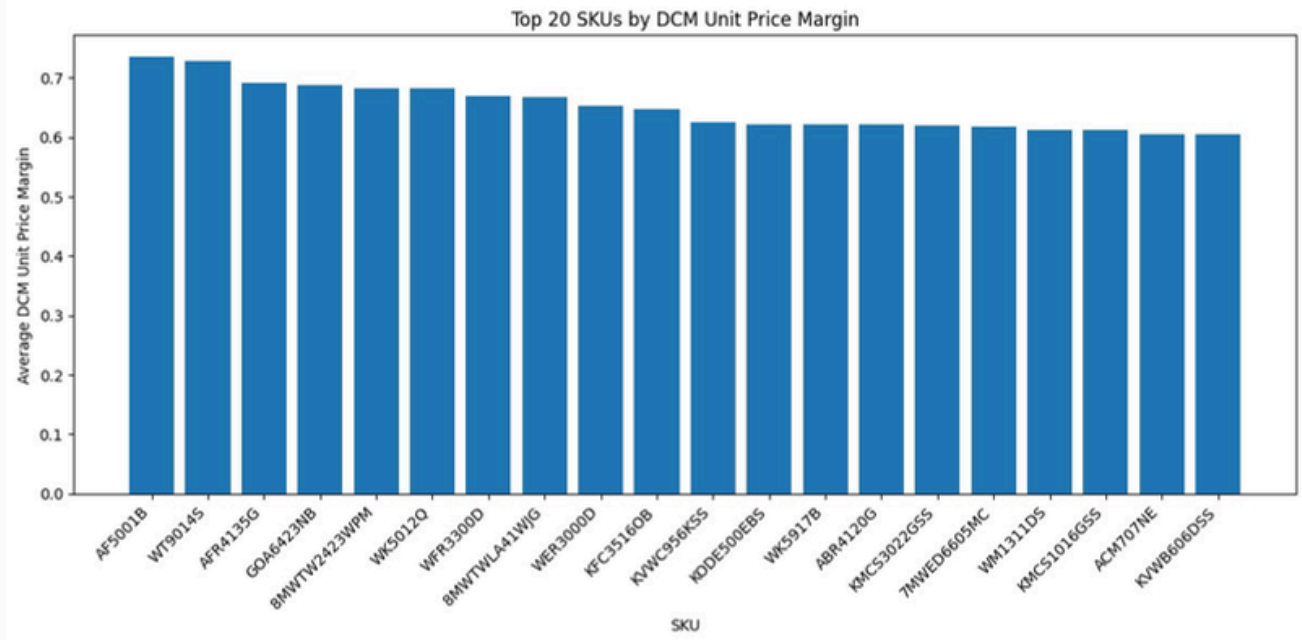


Trading partner with Biggest Volume

Coppel is the trading partner with the **highest sales value**, accounting for **17.1%** of the **total quantity sold**, followed by Elektra (14.4%) and D2C (13.5%). The 'Others' group collectively represents the largest overall share at 24.3%, combining several smaller trading partners.



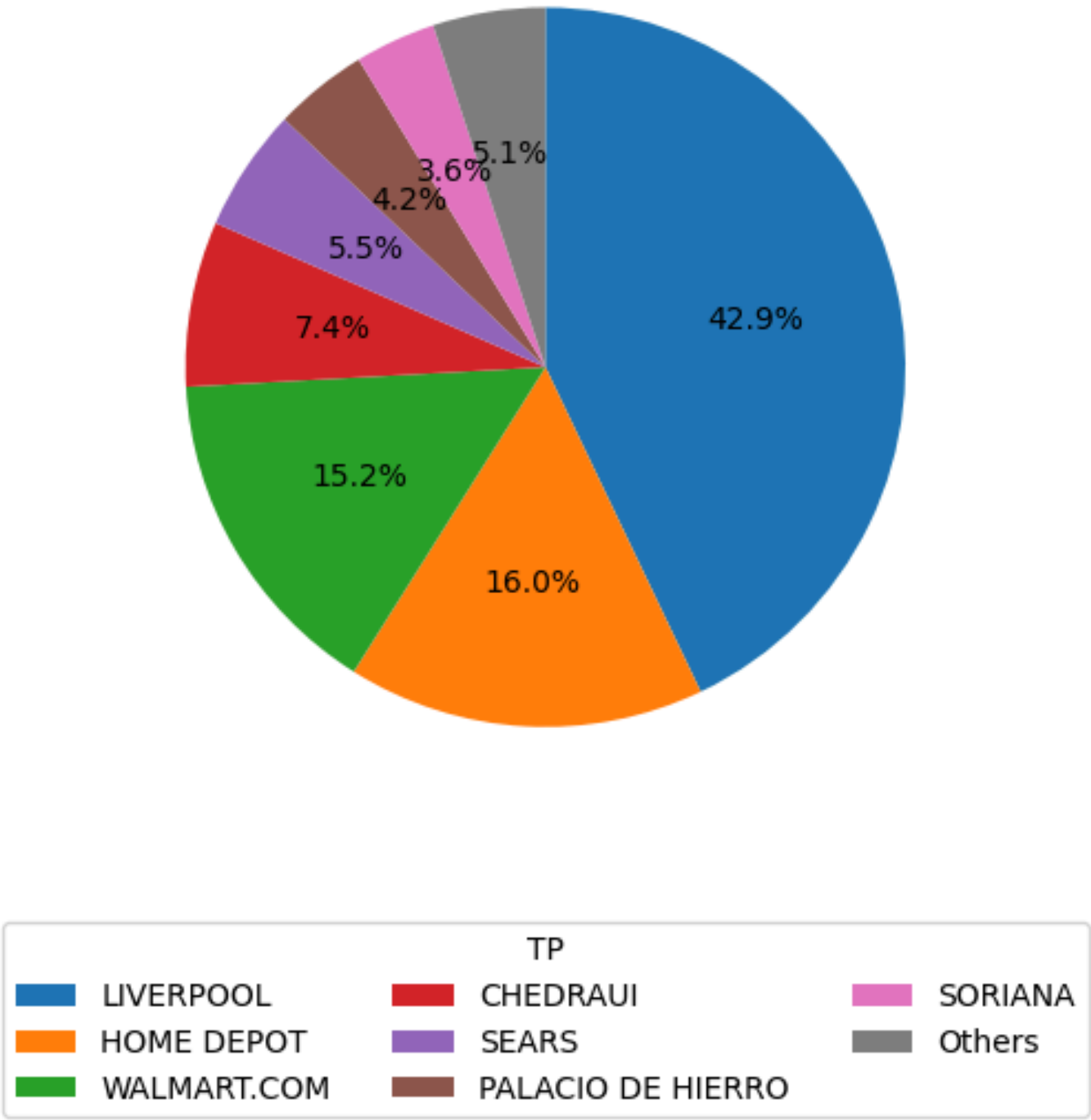
Top Sales & Price Leaders



	SKU	DCM_unit_price_margin
229	AF5001B	0.735016
1316	WT9014S	0.728410
262	AFR4135G	0.691525
444	GOA6423NB	0.688639
213	8MWTW2423WPM	0.682603

The SKU that sells for the highest unit price is AF5001B, with a of 1.06 billion.

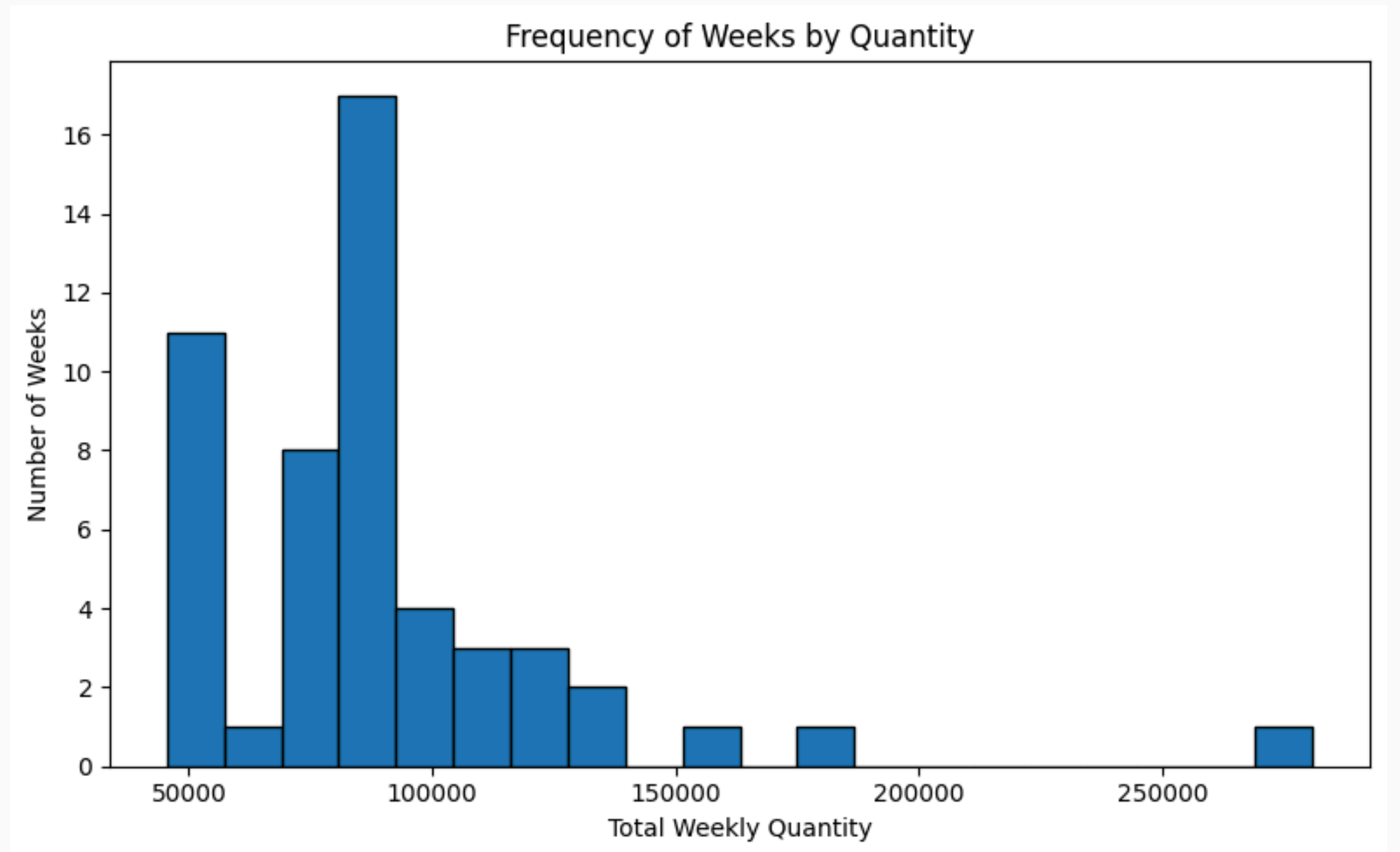
Trading Partner Distribution by Gross Sales



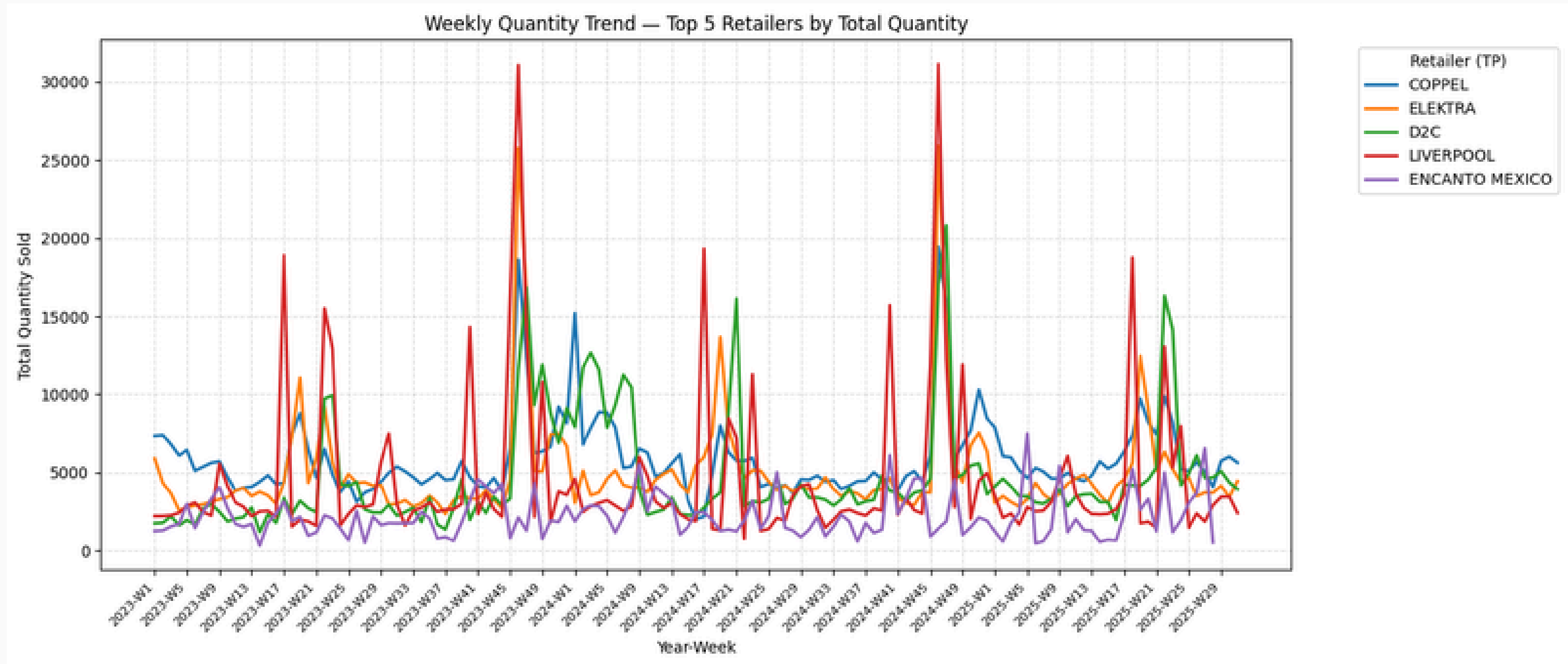
The trading partner with the highest sales (considering price) is LIVERPOOL with total sales of 5,720,360,806.08.

Distribution of Weeks by Total Quantity

Distribution shape:
Strongly right-skewed (few huge weeks dominate).
Main insight: Most weeks have moderate quantity, but rare extreme peaks contribute disproportionately to total quantity.



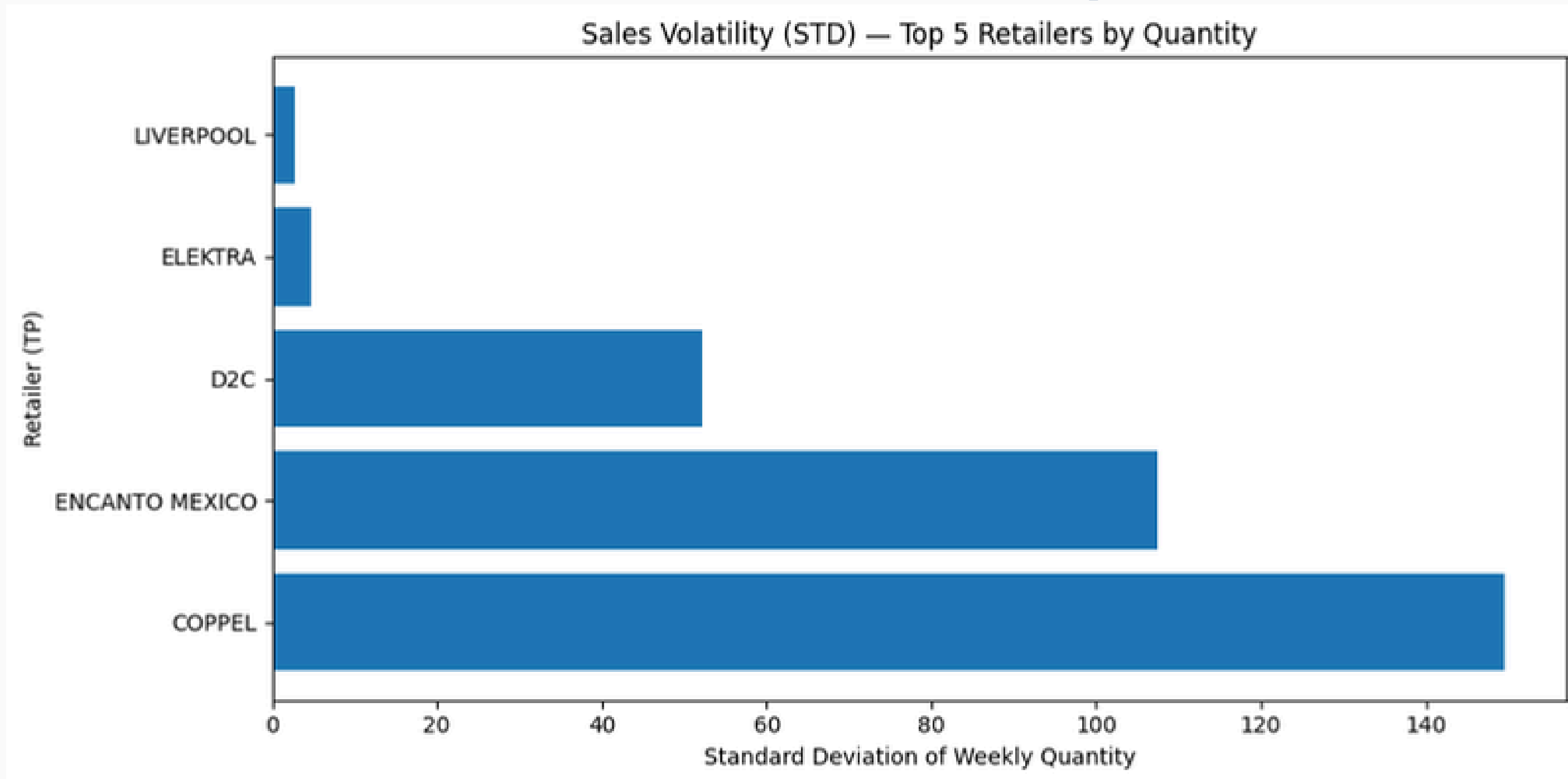
Weekly Trend



The chart tracks the weekly sales quantity for the top 5 retailers (Coppel, Elektra, D2C, Liverpool, and Encanto México) over time.

Each line represents how total units sold changed week by week during 2023–2025.

Sales Volatility



Coppel and Encanto Mexico have the most unpredictable weekly sales, while Liverpool and Elektra are very steady.

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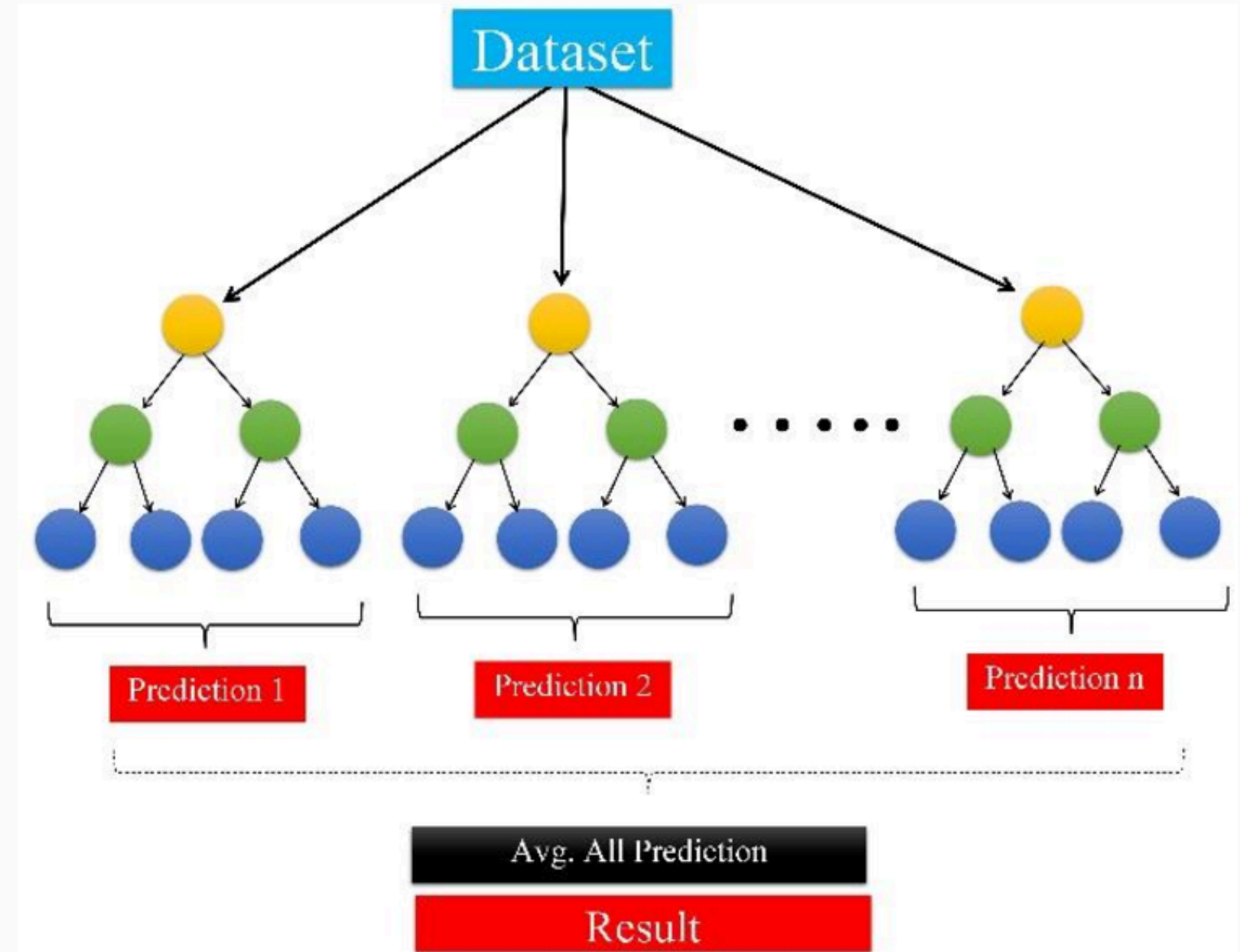
Machine Learning Models



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The Model

- Models Tested:
 - XGBoost Regressor – handles nonlinear relationships between price, cost, and sales.
 - Prophet – captures trend and seasonality.
 - Elasticity (log–log regression) – estimates price sensitivity.
- Train/Test Split: 70/30 (chronological order)



PROPHET

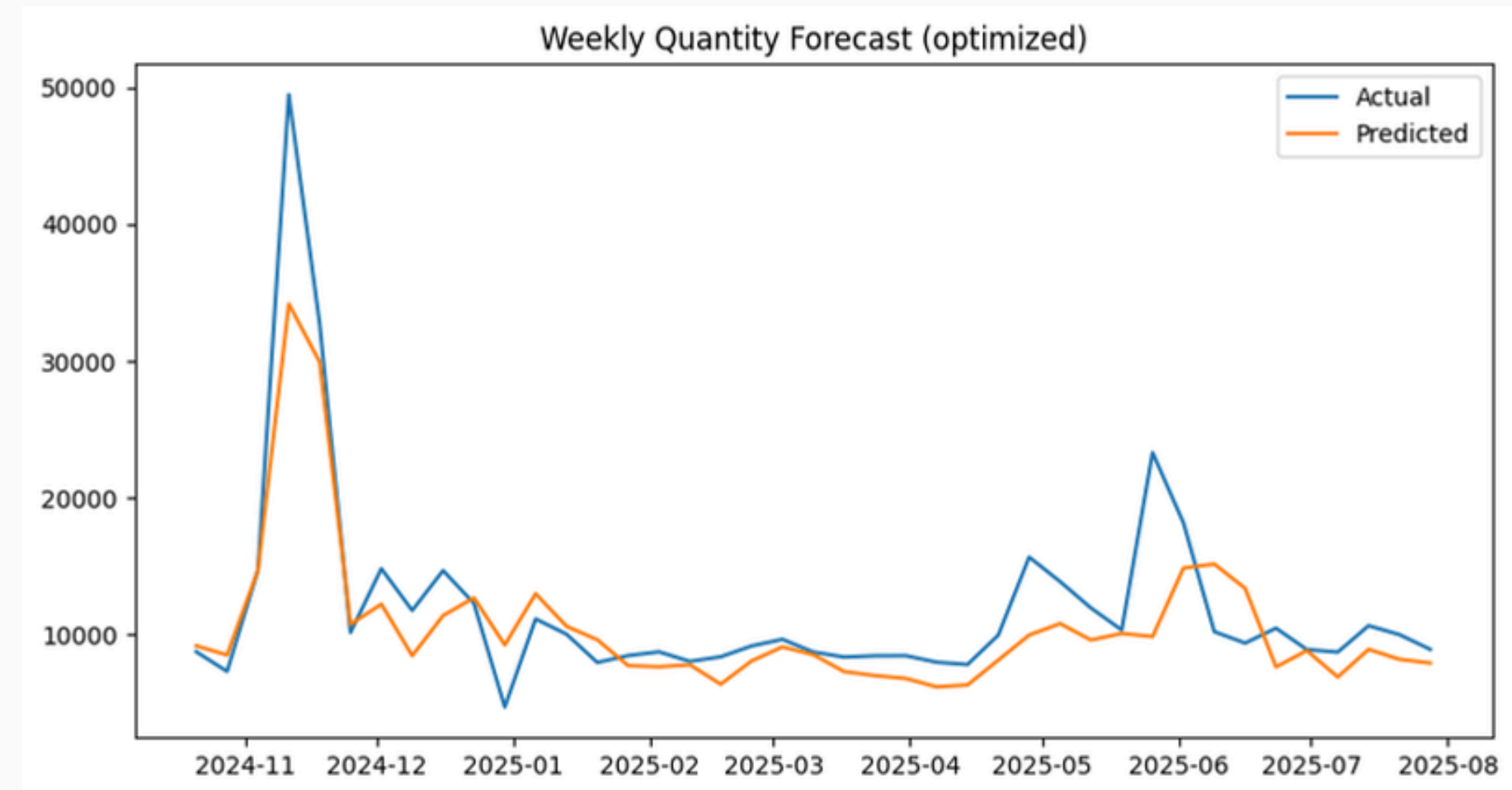
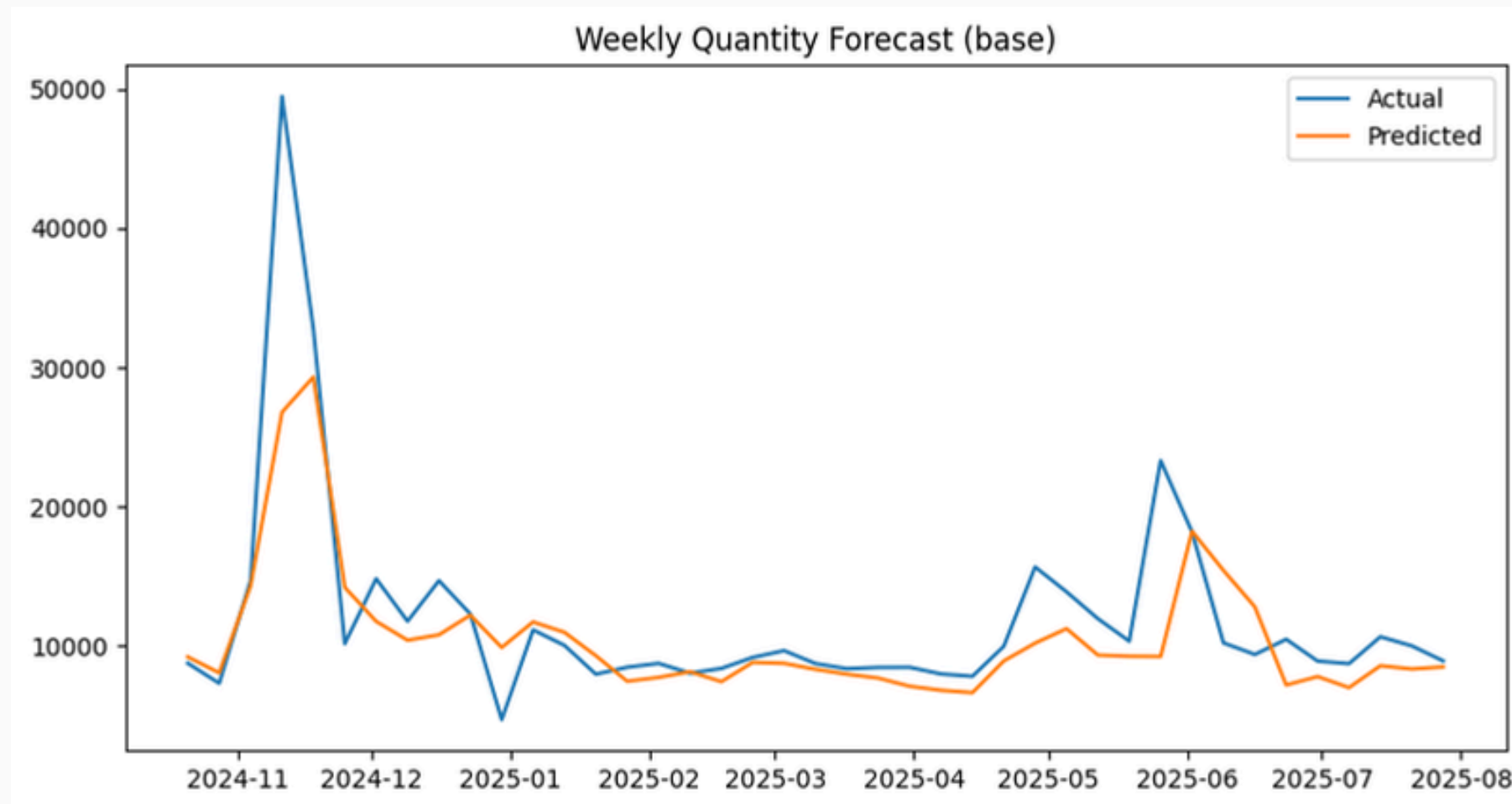
Model tuning

- For XGBoost: key parameters tuned using GridSearchCV included `n_estimators`, `learning_rate`, `max_depth`, `subsample`, and `colsample_bytree` to balance model complexity, accuracy, and overfitting risk.
- Tuning methods: GridSearchCV for XGBoost and manual adjustment for Prophet.

```
# Define hyperparameter grid
param_grid = {
    #boosting parameters
    'n_estimators': [100, 300, 500, 800, 1000],
    'learning_rate': [0.01, 0.05, .1, .15, .2, .25, .3],
    #tree parameters
    'max_depth': [3, 5, 7, 10],
    #sample parameters
    'subsample': [.5, .7, 1],
    'colsample_bytree': [.5, .7, 1]
}
```

```
Best parameters: {'colsample_bytree': 0.7, 'learning_rate': 0.3, 'max_depth': 3, 'n_estimators': 100, 'subsample': 0.5}
```


XGBoost Regressor base vs optimized



Initial XGBoost:

- MAE: 2530.975
- R2: 0.611
- RMSE: 4727.141
- MAPE: 19.039%

Features:

'week_sin', 'week_cos',
'price_avg_last_week',
'qty_last_week',
'qty_avg_prev_4w', 'event_effect'

Tuned XGBoost (optimized):

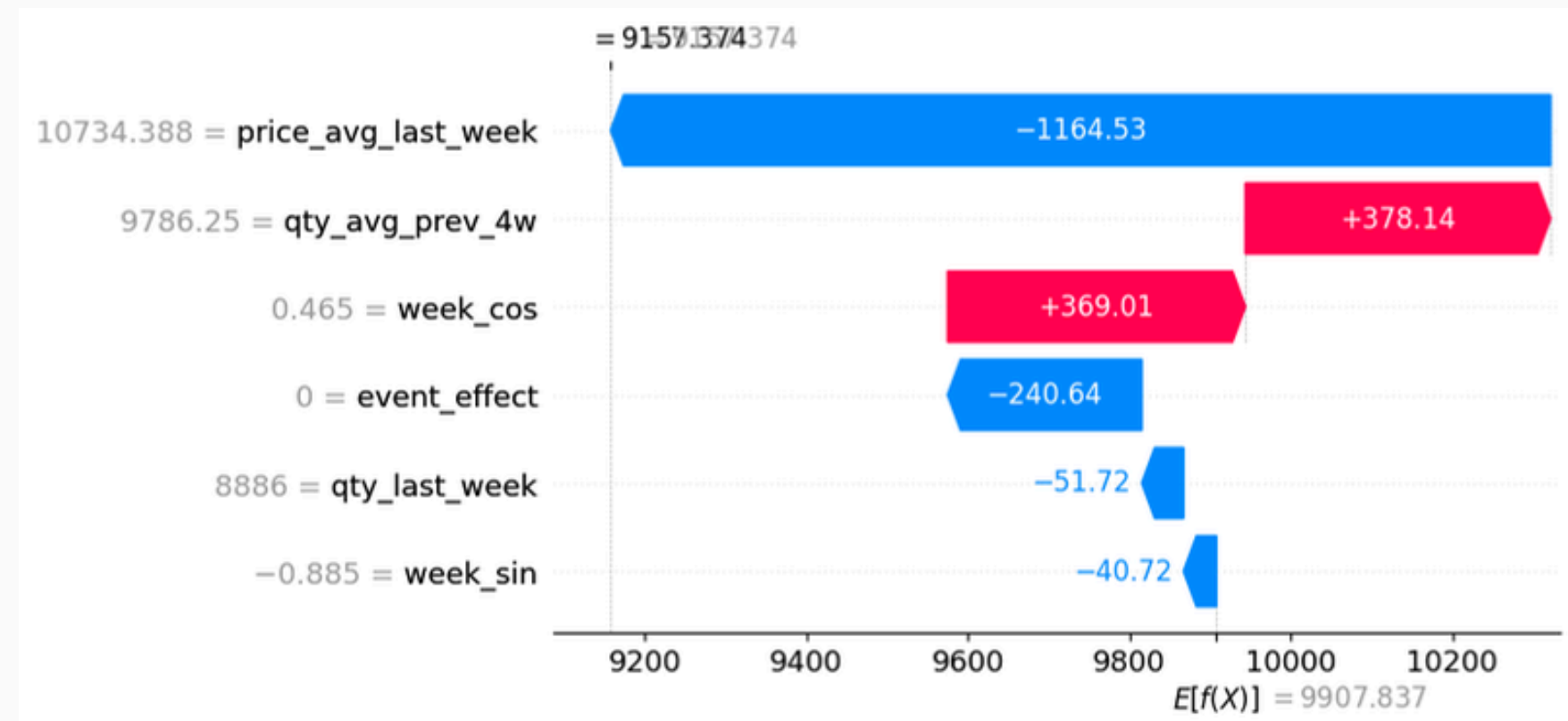
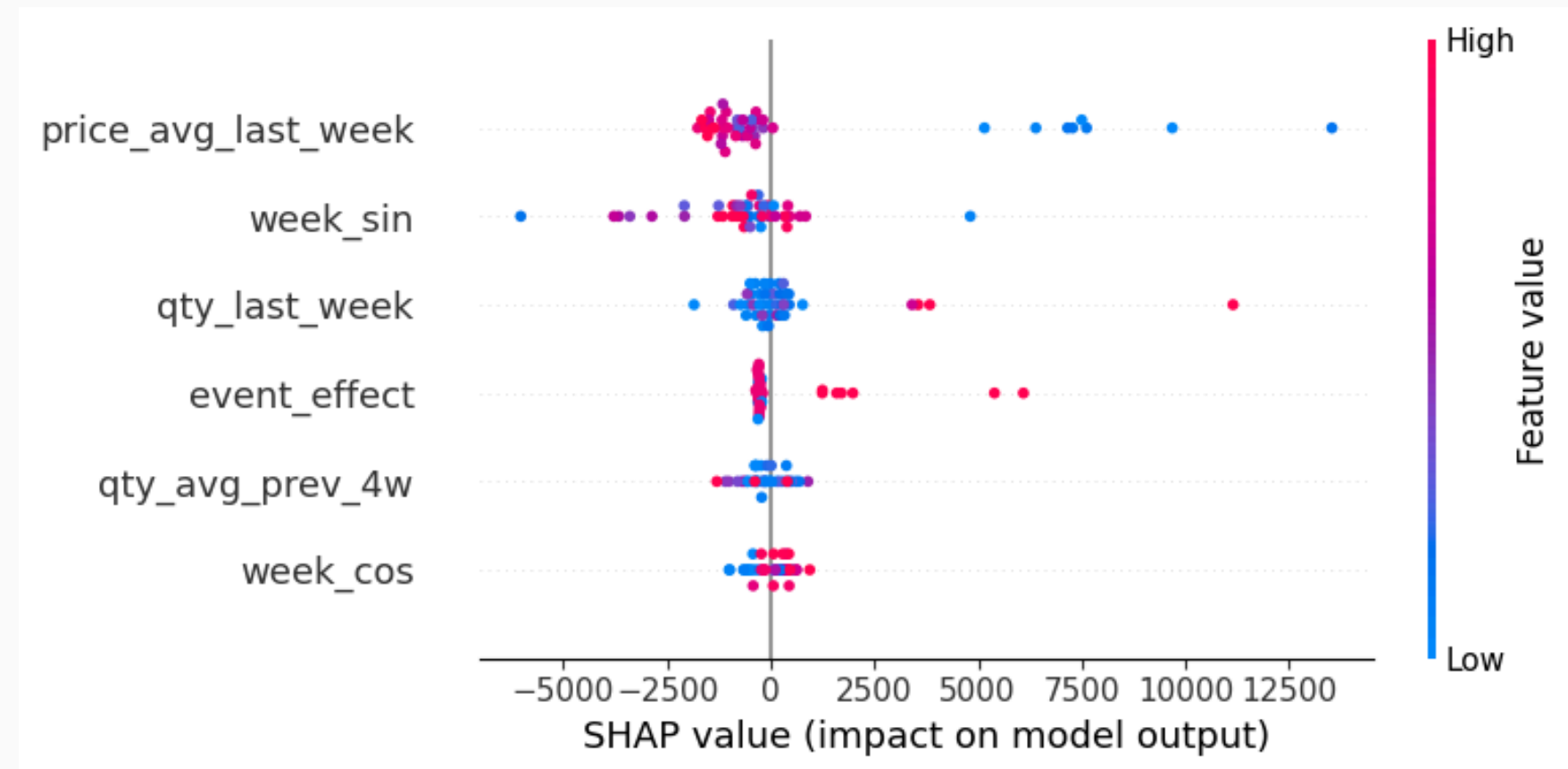
- MAE: 2443.156
- R2: 0.736
- RMSE: 3892.857
- MAPE: 19.498%

SHAP

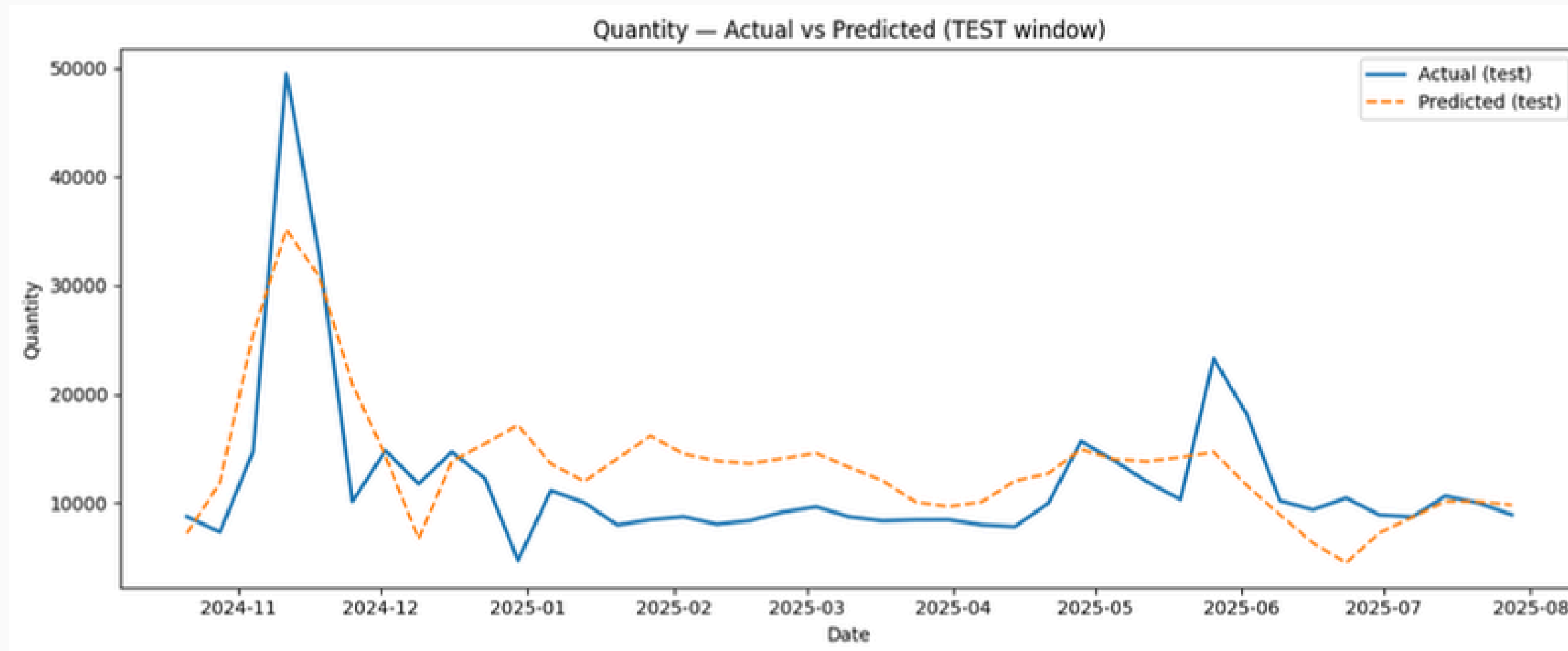
SHAP values explain how each feature influences the model's predictions by quantifying its contribution.

The most interesting points are:

- Last week's price strongly reduced the predicted quantity which is offset by last 4 weeks quantity with raise the predicted quantity.
- The absence of an event reduced the predicted quantity (which is intended).



Prophet



Features:

- Unit Price last week
- Quantity last week
- Quantity average last 4 weeks
- Event effect

Prophet model:

- $R^2 = 0.51$
- MAE = 4,060
- MAPE $\approx 40\%$

Model Performance

Initial XGBoost (manual hyperparameters): • MAE: 2530.975 • R2: 0.611 • RMSE: 4727.141 • MAPE: 19.039% • Baseline model	Tuned XGBoost (optimized): • MAE: 2443.156 • R2: 0.736 • RMSE: 3892.857 • MAPE: 19.498% • Major improvement after optimizing
Prophet model: • $R^2 = 0.51$ • MAE = 4,060 • MAPE $\approx$ 40% • RMSE: 5322.613 • Captures trend and seasonality • Less accurate for short-term variations	

Elasticity Analysis

$$\ln(Q_t) = \beta_0 + \beta_1 \ln(P_t) + \beta_2 \ln(C_t) + \beta_3 \ln(P_{comp,t}) + \dots + \epsilon_t$$

- We estimated price elasticity using a log-log regression model.
- If the coefficient on `ln_price` is -7 , that means a 1% increase in price leads to roughly a 7% decrease in quantity sold.
- Elasticity varied by segment: Top Load and Twin Tub showed the highest sensitivity, while Single Tub was the most stable.

segment	own_elasticity	cross_elasticity	r2	MAE	RMSE	MAPE_%
FRONT LOAD	-5.981902	1.097425	0.137132	4059.915348	5279.289005	36.75633
LAUNDRY CENTER	-7.298053	0.025389	0.241617	4059.915348	5279.289005	36.75633
SINGLE TUB	-3.417736	-0.781299	0.056416	4059.915348	5279.289005	36.75633
TOP LOAD	-8.634484	-2.706635	0.285064	4059.915348	5279.289005	36.75633
TWIN TUB	-8.77361	-3.850777	0.229583	4059.915348	5279.289005	36.75633

Profit-maximizing price

- Modeling at the segment level gives us a more stable and interpretable elasticity estimate.
- For instance, if a segment has an elasticity of -5 and an average cost of 6,600, the model suggests an optimal price around 8,250.

$$P^* = \frac{|\epsilon|}{|\epsilon| - 1} \times \text{Cost}$$

segment	own_elasticity	cross_elasticity	avg_cost	avg_price	opt_price
FRONT LOAD	-5.981902	1.097425	8099.619728	11749.541504	9725.428564
LAUNDRY CENTER	-7.298053	0.025389	15247.299506	20170.965607	17668.253848
SINGLE TUB	-3.417736	-0.781299	2234.923781	4163.306153	3159.310786
TOP LOAD	-8.634484	-2.706635	6646.488729	10394.45507	7517.076526
TWIN TUB	-8.77361	-3.850777	3091.141441	5403.506453	3488.786995

Results and interpretation

The tuned XGBoost model delivered the best performance, with `week_cos`, `qty_last_week`, and `week_sin` as the top predictive features. It achieved the lowest MAE and RMSE and the highest R^2 , showing strong accuracy and variance explanation. All models surpassed the simple average baseline, and XGBoost outperformed Prophet.



Prediction

```
def predict_quantity(week_number, tp, segment, price_last_week, qty_last_week, qty_avg_prev_4w):
    week_sin = np.sin(2 * np.pi * week_number / 52)
    week_cos = np.cos(2 * np.pi * week_number / 52)
    event_effect = df.loc[df['WEEK'] == week_number, 'event_effect'].iloc[0]
    x = np.array([[week_sin, week_cos, price_last_week, qty_last_week, qty_avg_prev_4w, event_effect]])
    y_hat = float(best_xgb.predict(x))

    quantity_tp = float(y_hat * df_sum.loc[df_sum['TP'] == tp, '%'].iloc[0])
    quantity_tp_segment = float(quantity_tp * segment_percent[tp].loc[segment_percent[tp]['segment'] == segment, '%'])

    result = pd.DataFrame([({
        'week_number': week_number,
        'tp': tp,
        'segment': segment,
        'price_last_week': price_last_week,
        'qty_last_week': qty_last_week,
        'qty_avg_prev_4w': qty_avg_prev_4w,
        'Overall quantity': y_hat,
        f'Quantity for {tp}': quantity_tp,
        f'Quantity for {tp} for {segment}': quantity_tp_segment
    })])

    return result
```

We are assuming that the distribution sales for TP and segment will maintain in the future.

Prediction

Event Scenarios:

- Buen Fin (Week 45):
 - Overall predicted quantity $\approx 32,762$
 - LIVERPOOL: 4,917
 - TOP LOAD segment: 3,180
- Día de las Madres (Week 18):
 - Overall predicted quantity $\approx 17,305$
 - ELEKTRA: 2,675
 - SINGLE TUB segment: 704
- Low Demand (Week 2):
 - Overall predicted quantity $\approx 5,775$
 - COPPEL: 1,451
 - SINGLE TUB segment: 476



Limitations and biases

- XGBoost: very accurate, but hard to understand how it decides.
- Prophet: good for trends, but not for sudden changes.
- Elasticity: assumes price effects always stay the same.
- Some products had little data.
- Models need to be updated often.

LIMITATION



Thank You